

THEJUS MAHAJAN PH. D.

Bioinformatics & Data Scientist



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CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in Biological Systems Modeling and High-Dimensional Data Analysis. Experienced during post-doc in applying stochastic population models in hindcast and forecast projections. Trained in Python and R pipelines for noisy data analysis. Seeking for a career shift to Bioinformatics. I am leveraging on my recent Bioinformatics and Biostatistics training along with the advanced mathematical modeling expertise gained during the Ph. D. and post-doc period to solve problems in human metabolic networks and clinical data analysis.

EXPERIENCE

Further Training – Bioinformatics

08/2025 – Present

CQ Beratung + Bildung, Berlin

Introduction to programming and Python, including Shell (BASH), programming methodology, and object-oriented programming in Python. Bioinformatics and databases, including NCBI, Biopython, web services, and database management with SQL. Applied bioinformatics, including sequence analysis and strategies, structural bioinformatics, software for structure visualization, and NGS analysis using Nextflow and Galaxy. Applied biostatistics, including statistical analysis using R, descriptive and multivariate statistics, ANOVA, PCA, HCA, workflows using R/Bioconductor, and Markov chain implementation. Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (GitHub).

Internship – Machine Learning in R

02/2026 – 04/2026

HealthTwiSt GmbH, Berlin

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,348 lines (–26.5%) with byte-identical output verified via `identical()`.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
- Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.

STRENGTHS

Advanced Scientific Modelling

Developed and implemented a novel biogeochemical model (CLC) within the GOTM-FABM framework.

Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector.

Bioinformatics

Successfully bridged physical science expertise with modern biological workflows.

SKILLS

Python	● ● ● ● ●
R (Tidymodels)	● ● ● ● ●
Nextflow	● ● ● ● ●
SQL	● ● ● ● ●
Fortran	● ● ● ● ●
LaTeX	● ● ● ● ●

LANGUAGES

English	● ● ● ● ●
German (B1)	● ● ● ● ●
French	● ● ● ● ●
Malayalam	● ● ● ● ●
Tamil	● ● ● ● ●

REFEREES

Dr. Andreas Busjahn

HealthTwiSt GmbH

andreas@busjahn.net

Prof. Dr. Kai Wirtz

Helmholtz-Zentrum Hereon

kai.wirtz@hereon.de

Hamburg, den 05.02.2026

- Created the `degirtools` R package (9 exported functions, roxygen2 docs, `devtools:::check()` passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
- Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

(Signature)
Thejus Mahajan

Guest Scientist

📅 05/2025 – 10/2025

Helmholtz-Zentrum Hereon, Geesthacht **Computational Biologist & Systems Modeler**

- Spearheaded a complex research project modeling evolutionary processes in marine environments, in collaboration with the Ecosystem Modeling Department.
- Applied advanced computational techniques to simulate the long-term impacts of environmental changes on marine ecosystems.

Post-doctoral Scientist

📅 08/2021 – 01/2025

University of Hamburg, Hamburg **Marine Ecosystem Modelling**

- Model development: Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework to project future climate warming scenarios.
- High-performance computing: Engineered models in Fortran and analyzed large NetCDF datasets using Python in a Linux environment.
- Statistical analysis and visualization of ecological time-series data using R.

Ph.D. Researcher Astrochemistry

📅 10/2015 – 09/2018

Univ. Paris-Saclay, France

- Big Data Analysis: Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- Algorithm Development: Utilized C++ for noise cancellation, signal smoothing.
- Simulation: Developed a molecular fragmentation model for the KIDA database.

EDUCATION

Ph.D. in Astrochemistry

2015 – 2018

Université Paris-Saclay, France

M.Sc. in Physics

2012 – 2015

National Institute of Technology Calicut, India

CGPA: 8.2/10